

Want:

Suppose L, K are compactly generated elementary groups and $K \curvearrowright X$ with comp/open point stabilizers (no need to be a faithful action). Let $U \in \mathcal{U}(L)$, then we have that

$$\gamma + \alpha \leq \mathcal{G}(L S_0(K, X))$$

where $\mathcal{G}(L) = \gamma + 2$, $\mathcal{G}(K) = \alpha + 2$

↳ First ideas:

1. Case K compact follows from the same argument as the other lemma

2. For the case K non-compact we will make an induction on γ and α , one after the other

3. Base case: $\gamma = 0$

$$\alpha < \mathcal{G}(K) \leq \mathcal{G}(L S_0(K, X))$$

4. Suppose that for all $\mathcal{Q}$ compactly generated non-trivial elementary group with $\mathcal{G}(\mathcal{Q}) < \gamma$ and for all $E \curvearrowright X$ elementary group with comp/op. pts stabilizers we have, for $U \in \mathcal{U}(\mathcal{Q})$

$$\delta + \beta < \mathcal{G}(\mathcal{Q} S_0(E, X))$$

where $\mathcal{G}(\mathcal{Q}) = \delta + 2$, $\mathcal{G}(E) = \beta + 2$

↑ (Induction hypothesis)

5 $\rightarrow$ Let L be s.t. $\mathcal{G}(L) = \delta + 2$ and K elementary with and compactly generated with $\mathcal{G}(K) = \alpha + 2$

6 $\rightarrow$ Because K is non-compact, let $Z \subset X$ be an infinite orbit

7 $\rightarrow$ Let $U \in \mathcal{U}(L)$ and define

$L \rightarrow$ was not used

$$N := \langle \langle U \rangle \rangle_L, H := N S_0(K, Z), J := \bigoplus_Z (N, U)$$

8 $\rightarrow$ H is closed in $L S_0(K, X)$, then

$$\mathcal{G}(H) \leq \mathcal{G}(L S_0(K, X))$$

9 $\rightarrow$ Because $N \trianglelefteq L$ we have $\text{Res}(L) \leq N$ and, because L is compactly generated

$$\delta + 1 = \mathcal{G}(\text{Res}(L)) \leq \mathcal{G}(N) \leq \mathcal{G}(L) = \delta + 2$$

10 $\rightarrow$ Let $(Q_i)_{i \in \mathbb{N}}$ be an increasing family of compactly generated groups with $0 \leq Q_i \leq Q_{i+1}$ and $\bigcup_{i \in \mathbb{N}} Q_i = N$. Then

$$\sup_{i \in \mathbb{N}} \mathcal{G}(Q_i) \leq \delta + 1 \rightarrow \text{This might not be true when } \mathcal{G}(N) = \delta + 2, \text{ so we cannot apply induction}$$

11 $\rightarrow$ Define $H_i := Q_i S_0(K, Z)$ and $\mathcal{G}(Q_i) = \beta_i + 2$. By the induction hypothesis we have $\beta_i + \alpha \leq \mathcal{G}(H_i)$

13 $\Rightarrow$ Then

$$\sup B_i + \alpha \leq \sup \mathcal{G}(H_i) \leq \sup \mathcal{G}(H)$$

$$\Rightarrow \gamma + \alpha \leq \sup(H) \leq \mathcal{G}(LS_0(K, X))$$

as desired.

~~For the next~~

The biggest problem I saw on this proof was the fact that we are not sure about the rank of N , so it is not possible to apply the induction

Also, at first I was trying to work with an idea similar to the other proof but I ended up not working with the $\text{Res}(H)$, which might be a problem

~~stop~~ It might be necessary to add some steps in between and maybe work inside $\text{Res}(L)$ ~~instead~~ instead of N

→ Second Try:

Same first 5 steps

6 → Let $R = \text{Res}(L) \Rightarrow \mathcal{G}(\text{Res}(L)) = \delta + 1$

7 → Let $(Q_i)_{i \in \mathbb{N}}$ be an increasing family of compactly generated open subgroups of R with $R = \bigcup_{i \in \mathbb{N}} Q_i \Rightarrow \mathcal{G}(R) = \sup^+ \mathcal{G}(Q_i)$ and $\mathcal{G}(Q_i) = \beta_i + 2, \beta_i < \delta$

8 → Apply the induction on

Let $U \in \mathcal{U}(R) \Rightarrow \exists m \in \mathbb{N} \text{ s.t. } \forall i \geq m, U \leq Q_i$

so, w.l.o.g, consider $U \leq Q_i \forall i$

9 → By the induction hypothesis

$$\beta_i + \alpha \leq \mathcal{G}(Q_i S_0(K, X))$$

10 → Then we have $(\bigcup_{i \in \mathbb{N}} Q_i S_0(K, X) = R S_0(K, X) \dots?)$

$$\sup \beta_i + \alpha \leq \sup_{i \in \mathbb{N}}^+ \mathcal{G}(Q_i S_0(K, X)) = \mathcal{G}(R S_0(K, X))$$

Would need to change induction

There we can have $\sup \beta_i = \delta$ or $\sup \beta_i = \delta + 1$, if we consider the case $\alpha \geq \omega$ we will always have the $+1$ "engulfed" by α

$\Rightarrow \delta + \alpha \leq \sup \beta_i + \alpha \leq \mathcal{G}(R S_0(K, X)) \rightarrow U \text{ might not be in } \mathcal{U}(L)$

↳ Third idea:

↳ Let $U \in \mathcal{U}(L)$ and $R := R_U(L)$ and $U_R := U \cap R$
(then U_R is open/compact in R with subspace topology) → False

↳ Observe that $RS_{U_R}(K, X)$ is closed on $LS_U(K, X)$, then

$$\mathcal{G}(RS_{U_R}(K, X)) \leq \mathcal{G}(LS_U(K, X)) \text{ compactly generated}$$

↳ Let $(Q_i)_{i \in \mathbb{N}}$ sequence of increasing open subgroups of R with $U_R \leq Q_i \forall i$ and $\bigcup_{i \in \mathbb{N}} Q_i = R$, let $\mathcal{G}(Q_i) = \beta_i + 2$

↳ By the induction hypothesis

$$\beta_i + \alpha \leq \mathcal{G}(Q_i S_{U_R}(K, X))$$

Then we have

$$\delta + \alpha \leq \sup \beta_i + \alpha \leq \mathcal{G}\left(\bigcup_{i \in \mathbb{N}} Q_i S_{U_R}(K, X)\right) = \mathcal{G}(RS_{U_R}(K, X))$$

completing the induction hypothesis.

↳ Because in our induction we say $\forall \delta < \delta$ and $\forall \alpha < \omega$,
we have $\delta + \alpha \leq \mathcal{G}(LS_U(K, X))$ ~~we don't need~~
~~that~~ the induction on α comes naturally

↳ Would like to work with $H \leq L$ ~~with~~ $U \leq H$
and $\mathcal{G}(H) = \delta + 1$ to work with the $HS_U(K, X)$ properly,

Observe that if L is compactly generated and $U \in \mathcal{U}(L)$ then
 $\hookrightarrow \mathcal{G}(L) \geq 3$

$U \times \text{Res}(L) \twoheadrightarrow U \text{Res}(L)$, then $\mathcal{G}(U \text{Res}(L)) = \mathcal{G}(\text{Res}(L))$

$\hookrightarrow$ Fourth idea

$\xrightarrow{6}$ Let $U \in \mathcal{U}(L)$, $R_\bullet := U \text{Res}(L)$

$\xrightarrow{7}$ Observe that $RS_\bullet(K, X)$ is closed on $LS_\bullet(K, X)$, then

$$\mathcal{G}(RS_\bullet(K, X)) \leq \mathcal{G}(LS_\bullet(K, X))$$

$\xrightarrow{8}$ Let $(Q_i)_{i \in \mathbb{N}}$ sequence of increasing open compactly generated subgroups of R with $U \subseteq Q_i \forall i$ and $\bigcup_{i \in \mathbb{N}} Q_i = R$, and let $\mathcal{G}(Q_i) = \beta_i + 2 \rightarrow$ Because $\mathcal{G}(Q) = \mathcal{G}(R) = \gamma + 1$ we have $\beta_i < \gamma$

$\hookrightarrow$ By the induction hypothesis

$$\beta_i + \alpha \leq \mathcal{G}(Q_i S_\bullet(K, X))$$

then we have

$$\gamma + \alpha \leq \sup \beta_i + \alpha \leq \mathcal{G}\left(\bigcup_{i \in \mathbb{N}} (Q_i S_\bullet(K, X))\right) = \mathcal{G}(RS_\bullet(K, X))$$

completing the induction hypothesis.

$\hookrightarrow$ I believe that if my induction was right then so is this proof, and it might be simpler than the other